EE/CS 172/130
Digital Logic and Design

Topic 7 – Practical Considerations in a Logic Circuit

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NAND / NOR Implementation

- NAND and NOR gates are called universal gates because they can be used to implement all basic logic gates (AND, OR, and NOT).
- They are easy to fabricate, cost-effective, and widely used in digital circuits.

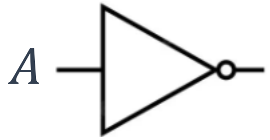
Steps for NAND/NOR Gate Implementation

1. Write the Boolean expression.
2. Simplify the expression
3. Convert to NAND/NOR form using equivalent NAND/NOR form for each gate
4. Simplify the NAND/NOR circuit by cancelling NAND/NOR inverters in series.
5. Draw the final logic diagram
6. Verify the implementation by making a truth table or simulating the circuit to ensure it matches the original logic function.

NAND Implementation

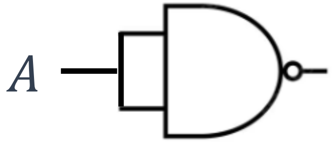
Gate	NAND Equivalent	NOR Equivalent

Gate



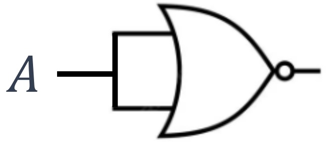
$\bar{A}$

NAND Equivalent

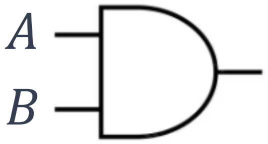


$\overline{A.A} = \bar{A}$

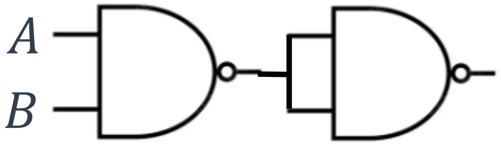
NOR Equivalent



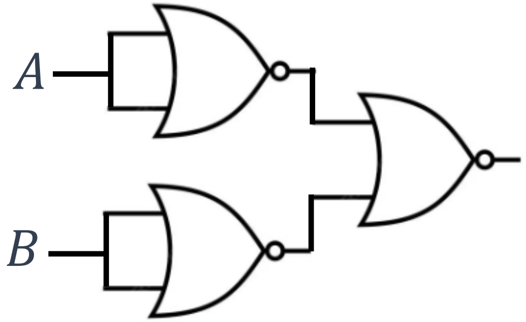
$\overline{A + A} = \bar{A}$



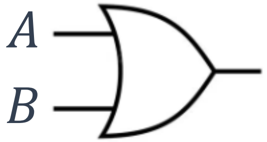
$A.B$



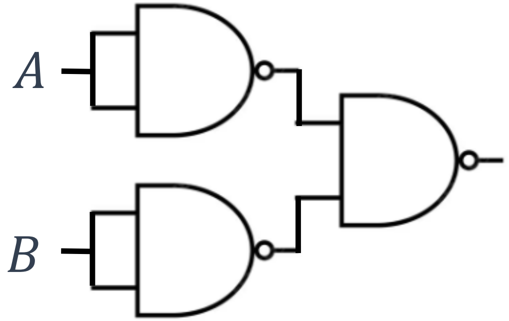
$\overline{\overline{A.B}}$
$= A.B$



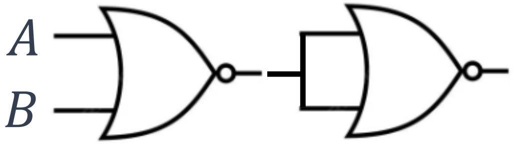
$\overline{\bar{A} + \bar{B}}$
$= \bar{\bar{A}}.\bar{\bar{B}}$
$= A.B$



$A + B$



$\overline{\bar{A}\bar{B}}$
$= \bar{\bar{A}} + \bar{\bar{B}}$
$= A + B$



$\overline{\overline{A + B}}$
$= A + B$

Practice Problems

- Give NAND and NOR equivalent circuits for the following expressions.
  - $F = \bar{A}B + A\bar{B}$
  - $F = (\bar{A} + B)(A + \bar{B})$

Integrated Circuits

- An Integrated Circuit (IC) is an electronic circuit consisting of active and passive components (such as transistors, resistors, capacitors) fabricated on a single semiconductor chip and enclosed in an individual package.
- Technically, a chip refers to the actual piece of silicon on which the circuit is fabricated. However, in practice, the terms chip and package are often used interchangeably.

Categories of ICs (Based on Number of Gates)

Category	Number of Gates	Examples
SSI (Small Scale Integration)	< 12 gates	Individual gates, flip-flops, memory elements
MSI (Medium Scale Integration)	12 – 100 gates	Functional blocks like decoders, multiplexers, counters
LSI (Large Scale Integration)	100 – 1,000 gates	Subsystems such as memory sections, arithmetic units
VLSI (Very Large Scale Integration)	1,000 – 1,000,000 gates	Microprocessors, large memories, complex controllers
ULSI (Ultra Large Scale Integration)	> 1,000,000 gates	Modern microprocessors, SoCs
GLSI (Giga Large Scale Integration)	≈ 1 billion gates	High-end CPUs, GPUs, advanced AI accelerators

IC Technologies

- The two prime technologies used to implement digital integrated circuits (ICs) are:
  - TTL (Transistor-Transistor Logic)
  - CMOS (Complementary Metal-Oxide Semiconductor)

TTL (Transistor-Transistor Logic)

- Built using bipolar junction transistors (BJTs).
- Known for fast switching speed but higher power consumption.

Input Voltage Levels

- Logic LOW: 0 V – 0.8 V
- Logic HIGH: 2 V – 5 V
- Region between 0.8 V and 2 V is undefined.

Output Voltage Levels

- Logic LOW: Typically ≤ 0.4 V
- Logic HIGH: Typically ≥ 2.4 V (but never reaches full Vcc in standard TTL)
- No undefined region for output; output is guaranteed to stay within spec if loading limits are met.

Float High Characteristic

- An unconnected input floats to logic HIGH by default due to internal biasing current.

TTL (Transistor-Transistor Logic)

Typical IC Series

- 74 / 54 Series
  - 74xx - Commercial (0°C to +70°C)
  - 54xx - Military / Industrial (-55°C to +125°C)

- Subfamilies (improved versions):
  - 74LSxx – Low Power Schottky (lower power, faster than standard TTL)
  - 74ALSxx – Advanced Low Power Schottky (even lower power)
  - 74Fxx – Fast TTL (higher speed)
  - 74Sxx – Schottky TTL (fast, but higher power dissipation)

CMOS (Complementary Metal-Oxide Semiconductors)

- Built using metal-oxide semiconductor field effect transistor (MOSFETs).
- Known for very low power consumption and very high integration density (used in almost all modern ICs, including processors).

Voltage Levels (For standard 5V CMOS logic)

- Logic LOW (0): 0 – 1.5 V
- Logic HIGH (1): 3.5 – 5 V
- Region between 1.5 V and 3.5 V is undefined.

- Floating inputs are dangerous in CMOS as they can cause unpredictable outputs and high power dissipation. Always tie them HIGH or LOW.

CMOS (Complementary Metal-Oxide Semiconductors)

Typical IC Series

- 4000/4500 Series CMOS
  - Operates on a wide voltage range (3V–15V).

- 74Cxx Series
  - CMOS versions of TTL 74xx series (pin-compatible, but slower).
- 74HCxx (High-Speed CMOS)
  - Faster CMOS series, same pin configuration as TTL.
- 74HCTxx (High-Speed CMOS, TTL-Compatible)
  - Same as 74HCxx but with TTL-compatible input voltage levels, so they can directly interface with TTL logic.

- TTL → historically dominant, faster but more power-hungry.
- CMOS → modern choice, low power, high density, used in nearly all VLSI/ULSI circuits (microprocessors, memories, SoCs).

Propagation Delays in Gates

- The time taken for a change in input of a logic gate to cause the corresponding change at its output.

Typical Range for Technologies

- TTL gates: propagation delay $\approx$ 10 ns (typical for standard TTL).
- CMOS gates: can be faster or slower depending on design; modern CMOS can be < 1 ns.

Active Low Logic

- Active-low logic means that a signal, device, or function is considered **active** (enabled, true, or asserted) when the voltage level is LOW (logic 0), and inactive when the voltage is HIGH (logic 1).

	Active High	Active Low
1	ON	OFF
0	OFF	ON

- Active-low signals are usually indicated by a bar/overline on the signal name (e.g., $\overline{RESET}$, $\overline{EN}$, $\overline{CS}$), or by a small bubble with the signal line.

- Widely used in digital circuits because TTL inputs naturally float HIGH, so pulling them LOW is easier and more reliable for enabling functions.

Tri-State ICs

- Tri-state (or three-state) logic devices have three output states:
  - Logic HIGH (1)
  - Logic LOW (0)
  - High Impedance (Hi-Z) → electrically disconnected state

- Allows multiple outputs to share a common bus without interfering with each other.
- The Hi-Z state removes the IC from the circuit so another device can drive the line.

- Example Devices: Bus drivers, memory outputs, microcontroller ports.